MIL-W-63329A (AR)
5 DECEMBER 1986

SUPERSEDING
MIL-W-63329 (AR)
30 May 1979

MILITARY SPECIFICATION

WARHEAD, GM, HE, M248
LOADING, ASSEMBLY AND PACKING

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Quality Assurance Provisions and special requirements for the loading, assembly and packing of one type of warhead, designated as Warhead, GM, HE, M248.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1336
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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection for Attributes
MIL-STD-00453 - Inspection, Radiographic

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9312583 - Warhead, GM (HE) M248, Assembly, Loading and Marking Diagram.
9340599 - Packaging for Box, Wood for Warhead, GM (HE): M248 or M248 Inert Parts Assembly

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications and standards.

3.2 Components and assemblies. Components and assemblies shall comply with all the requirements specified on the drawings, specifications and standards.

3.3 Radiographic requirements. Warheads shall meet the radiographic inspection criteria specified in the subparagraphs and Table I.

3.3.1 Cracks. Cracks in the explosive charge are permissible provided a charge separation does not exist. A charge separation is defined as the formation of a multi-segment charge caused by a crack, or cracks, extending entirely through the charge in any direction.

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3.3.2 Cavities. Cavities within the explosive charge shall not exceed the areas or dimensions specified in Table I. Porous areas shall be treated as cavities, except that 80% of the projected length and 80% of the projected area shall be considered for acceptance purposes.

TABLE I

Radiographic requirements

	Segment A*	Segment B*
Sum of projected areas of all cavities (square inch)	1/2	1
Maximum projected length of any cavity (inch)	1/2	1

*Segments are identified in Figure 1.

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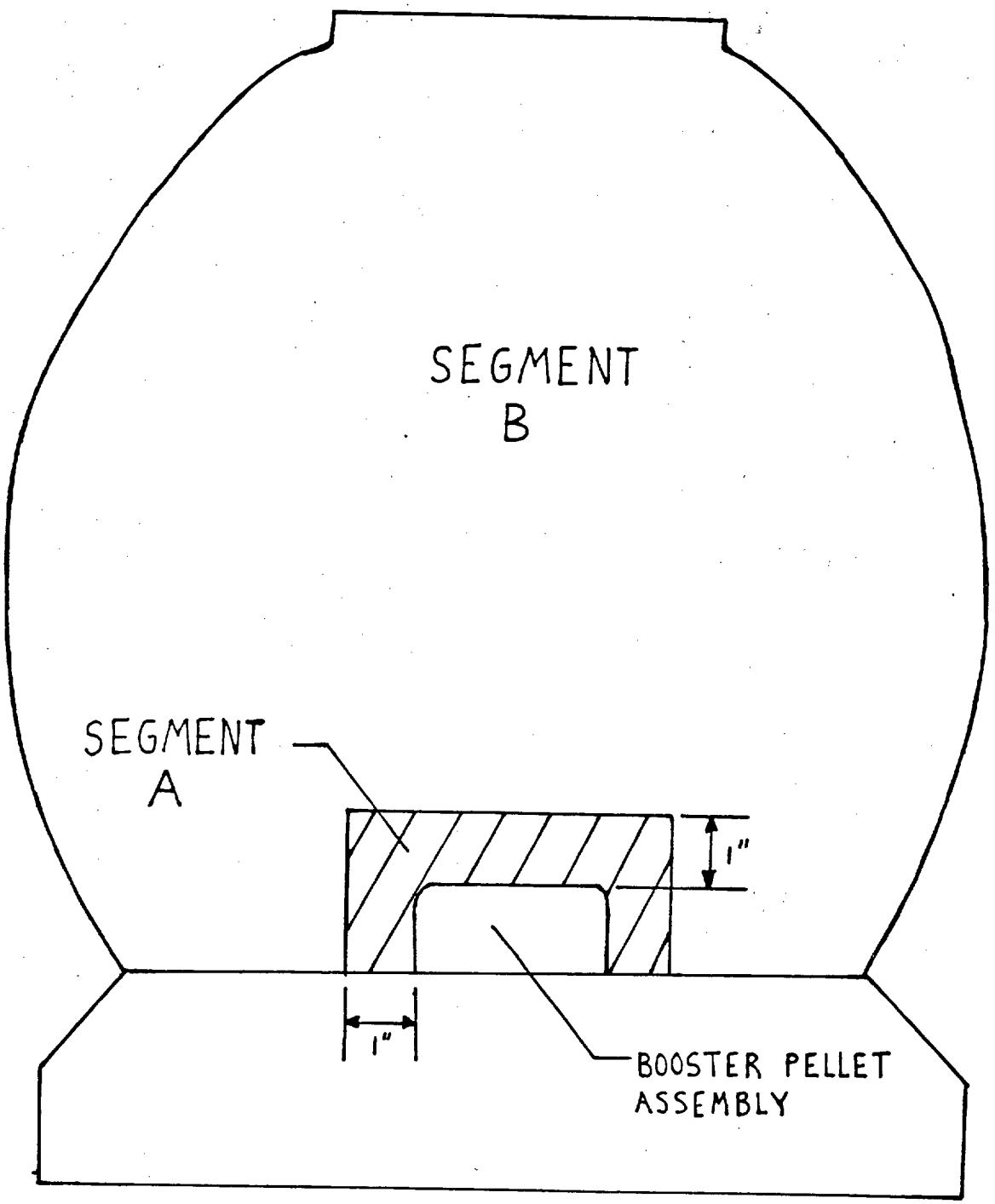


Figure 1

Patriot Warhead Radiographic Examination

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3.3.3 Cracks or laminations of booster pellets. Booster pellets shall be free of cracks or laminations.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article shall be as specified in the contract.

3.5 Workmanship. All warheads shall be loaded in a thorough manner. Warhead assemblies, kettles and related pouring apparatus shall be free of dirt, grease, rust, corrosion products or other foreign material. Care shall be taken during loading or subsequent probing of the explosive cast that the thin-walled retaining cup is not damaged by a probe or paddle of any kind. All exterior warhead surfaces shall be free of explosive, and attentive care shall be taken during loading, inspection and handling to prevent nicking and scoring of the aluminum base rim and end. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplied and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Booster Pellet	9280766	5
Cup Booster	9322124	5
Booster Disc	9280641	5
Spacer	9312516	5 ea size
Booster Cup Assembly (HE)	9312721	5
Retaining Ring	9322130	5
Warhead GM (HE) M248 Loading and Marking Assy	9312583	5

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.2.1					See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Booster Pellet (Dwg. 9280766)				
	Examination for Defects	5		3.1, 3.2	4.4.2.1
	Cup, Booster (Dwg. 9322124)				
	Examination for Defects	5		3.2	4.4.2.2
	Booster Disc. (Dwg. 9280641)				
	Examination for Defects	5		3.2	4.4.2.3
	Spacer (Dwg. 9312516)				
	Examination for Defects	5 ea size		3.2	4.4.2.4
	Booster Cup Assembly (HE) (Dwg. 9312721)				
	Examination for Defects	5		3.2	4.4.2.5
	Radiographic Examination	5		3.3.3	See note

NOTES: Each Booster Cup Assembly (HE) shall be radiographed in accordance with MIL-STD-00453, quality level 2. Two radiographs shall be taken radially and 90 degrees apart, and one radiograph taken axially for each assembly.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH 4.3.2.1	TITLE		SHEET 2 OF 2		DRAWING NUMBER See below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Retaining Ring (Dwg. 9322120)				
	Thickness	5		3.2	
	Protective finish missing or improper	5		3.2	
	Warhead GM (HE) M248, Loading and Marking Assembly (Dwg. 9312583)				
	Examination for Defects	5		3.2, 3.3	4.4.2.6, 4.4.2.7, 4.4.2.8

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280766
4.4.2.1	Booster Pellet				NEXT HIGHER ASSEMBLY 9312721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Certification of material missing	per lot	-	3.1	Visual/See note
102	Pellet density, min.		0.40%	3.2	Gage
103	Pellet height		1.0%	3.2	Gage
104	Pellet diameter		1.0%	3.2	Gage
105	Evidence of a crack or lamination		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES: A certificate of conformance of the material to drawing specification requirements shall be available for verification.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9322124
4.4.2.2	Cup, Booster				NEXT HIGHER ASSEMBLY 9312721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.65%	3.2	Gage
Minor					
201	Wall thickness		1.0%	3.2	Gage
202	Length		1.0%	3.2	Gage
203	Corner radius maximum		4.0%	3.2	Gage
204	Protective finish missing or improper		1.0%	3.2	Visual
205	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280641
4.4.2.3	Booster Disc				NEXT HIGHER ASSEMBLY 9312721
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		2.5%	3.2	Gage
Minor					
201	Thickness		4.0%	3.2	Gage
202	Protective finish missing or improper		4.0%	3.2	Visual
203	Evidence of poor workmanship		4.0%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312516
4.4.2.4	Spacer (2 sizes)				NEXT HIGHER ASSEMBLY 9312583
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Certification of material missing	per lot	-	3.1	Visual/See note
Minor					
201	Diameter		4.0%	3.2	Gage
202	Thickness		4.0%	3.2	Gage
203	Evidence of poor workmanship		4.0%	3.5	Visual

NOTES: A certificate of conformance of the material to the drawing specification requirements shall be available for verification.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312721
4.4.2.5	Booster Cup Assembly (HE)				NEXT HIGHER ASSEMBLY 9312583
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spacer or disc missing		0.40%	3.2	Visual
102	Crimp not continuous over 360°		0.40%	3.2	Visual
103	Length of assembly, max		0.40%	3.2	Gage
104	Marking missing or improper		100%	3.2	Visual
105	Sealant missing or improper		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312583
4.4.2.6	Warhead GM (HE) M248, Loading and Marking Assembly (Prior to Loading)				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	CAUTION: Booster cup assembly shall not be installed until explosive is cast.				
Critical	None defined				
Major 101	Certificate of conformance for explosive missing	per lot	-	3.1	Visual/See note
Minor 201	Interior primer coat missing		1.0%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.5	Visual

NOTE: A certificate of conformance of the explosive to the drawing specification requirements shall be available for verification.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Warhead GM (HE) M248, Loading and Marking Assembly (Prior to assembly of nose plate and booster cup assembly)				9312583
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of explosive in nose threads or on surfaces mating with nose plate		100%	3.5	Visual
102	Gasket missing or damaged		0.40%	3.2	Visual
103	Sealant missing or damaged		0.40%	3.2	Visual
104	Retaining cup well, dented or damaged		100%	3.5	Visual
Minor					
201	Depth to explosive		0.65%	3.2	Gage
202	Spacer missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63329A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Warhead GM (HE) M248, Loading and Marking Assembly				9312583
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of explosive on any exterior warhead surface		100%	3.5	Visual
102	Nose plate torquing procedure or torque value improper		100%	3.2	Gage
103	Booster cup assembly improperly oriented or loose		0.40%	3.2	Visual/Manual
104	Radiographic examination		100%	3.3	Gage/4.5.1
105	Center of gravity		1.0%	3.2	Gage
106	Assembly weight		100%	3.2,3.3	Gage
Minor					
201	Marking missing or improper		1.0%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9340599
4.4.2.9	Packaging for Box, Wood for Warhead, GM (HE): M248 or M248 Inert Parts Assembly				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cushion missing or improper		1.0%	3.2	Visual
Minor					
201	Marking missing or improper		1.5%	3.2	Visual
202	Strapping missing or improper		1.5%	3.2	Visual
203	Base cover missing		1.5%	3.2	Visual
204	Evidence of poor workmanship		1.5%	3.5	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Radiographic examination. Each warhead shall be radiographed in accordance with MIL-STD-00453, Inspection, Radiographic, radiographic quality level 2. Two radiographs taken radially and 90 degrees apart, and one radiograph taken axially shall be required for each warhead. The radiographs shall then be examined to determine compliance with the radiographic inspection criteria for the warhead. Any warhead failing to meet the requirements shall be classed defective and removed from the lot.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with drawing 9340599.

5.2 Packing. The parts shall be packed in accordance with drawing 9340599.

5.3 Marking. The wood packing box shall be marked in accordance with drawing 9340599.

6. NOTES

6.1 Intended use. The warhead covered by this specification is intended for use on the PATRIOT missile.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-I (D), Dover, N.J. 07801-5001.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Subject term (key word) listing.

Warhead
Booster pellet
Radiographic inspection

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian
Army-AR

Preparing activity:
Army-AR

(Project: 1336-A491)

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